

■ Scrutor: Fault Detection Technical Report

AI-Powered Predictive Maintenance & Algorithm Guide | Deployed App:
<https://scrutor-alpha.vercel.app/>

1. Executive Summary & Technology Stack

Scrutor is a full-stack predictive maintenance application designed to monitor industrial machinery health. It takes live operational telemetry parameters (voltage, current, motor speed, temperature, vibration, ambient temperature, humidity) to deliver instant fault diagnostics and actionable maintenance directives.

Backend Framework: FastAPI (Python) running on Uvicorn with in-memory lifespan model loading.

Frontend Framework: Next.js 15 App Router, React 19, Tailwind CSS v4, React Hook Form, and Zod.

Cloud Deployment: Next.js SPA hosted on Vercel, FastAPI backend microservice hosted on Render.

2. Machine Learning Pipeline & Data Scaling Process

Core Classifier: *RandomForestClassifier(n_estimators=200)* from Scikit-Learn.

Feature Scaler: *StandardScaler* from Scikit-Learn.

Why Random Forest was selected:

- **Non-Linear Dynamics:** Industrial failures involve non-linear interactions between multiple parameters (e.g. combined thermal overload and mechanical vibration). Random Forest constructs 200 decision trees to capture these multi-variable failure boundaries cleanly.
- **Overfitting Prevention:** Bagging ensemble aggregation reduces model variance.
- **Probabilistic Outputs:** Supports *.predict_proba()* for confidence score calculation.

Internal Scaling Process:

1. **Physical Min-Max Normalization:** Raw real-world inputs (e.g. 220.5 V, 1500 RPM) are normalized into the [0.0, 1.0] range using physical operational bounds: Voltage (180-260V), Current (0-30A), Motor Speed (0-3000 RPM), Temp (0-120°C), Vibration (0-5g), Ambient Temp (-10 to 60°C), Humidity (0-100%).
2. **Z-Score Standardization:** Normalized inputs are passed into *scaler.transform()* to compute standard Z-scores: $z = (x - \mu) / \sigma$ before classification.

3. Test Telemetry Parameters for 4 Diagnostic States

Parameter	Normal	Warning	Worst Condition	Critical
Voltage (V)	220.0 V	251.1 V	232.0 V	191.6 V
Current (A)	6.7 A	11.1 A	7.7 A	6.1 A
Motor Speed (RPM)	1500 RPM	1311 RPM	1401 RPM	1114 RPM
Motor Temp (°C)	32.2 °C	106.1 °C	29.9 °C	97.7 °C
Vibration (g)	0.20 g	4.00 g	0.80 g	0.50 g
Ambient Temp (°C)	25.0 °C	39.0 °C	-1.7 °C	48.2 °C
Humidity (%)	35.0 %	10.8 %	65.3 %	10.4 %
Confidence	100.0%	93.0%	98.5%	99.5%

4. Physics & Engineering Explanations of Fault States

1. Normal State (Class 0):

Cool core temperature (32.2°C) and minimal vibration (0.20g) indicate smooth mechanical rotation with zero friction or mechanical wear. Current draw (6.7A) confirms baseline electrical operating efficiency.

2. Warning State (Class 1):

High mechanical vibration (4.00g) combined with high core temperature (106.1°C) indicates early-stage shaft misalignment, bearing degradation, or mechanical unbalance. Requires scheduled maintenance.

3. Worst Condition State (Class 2):

Sub-zero ambient temperature (-1.7°C) combined with high relative humidity (65.3%) creates lubricant condensation/thickening inside gearboxes, producing viscous drag and motor speed reduction (1401 RPM).

4. Critical State (Class 3):

Severe Under-Voltage (191.6V) deprives motor torque, causing rotation speed collapse (1114 RPM) while high ambient heat (48.2°C) prevents thermal dissipation. Poses severe thermal breakdown risk; requires emergency shutdown.